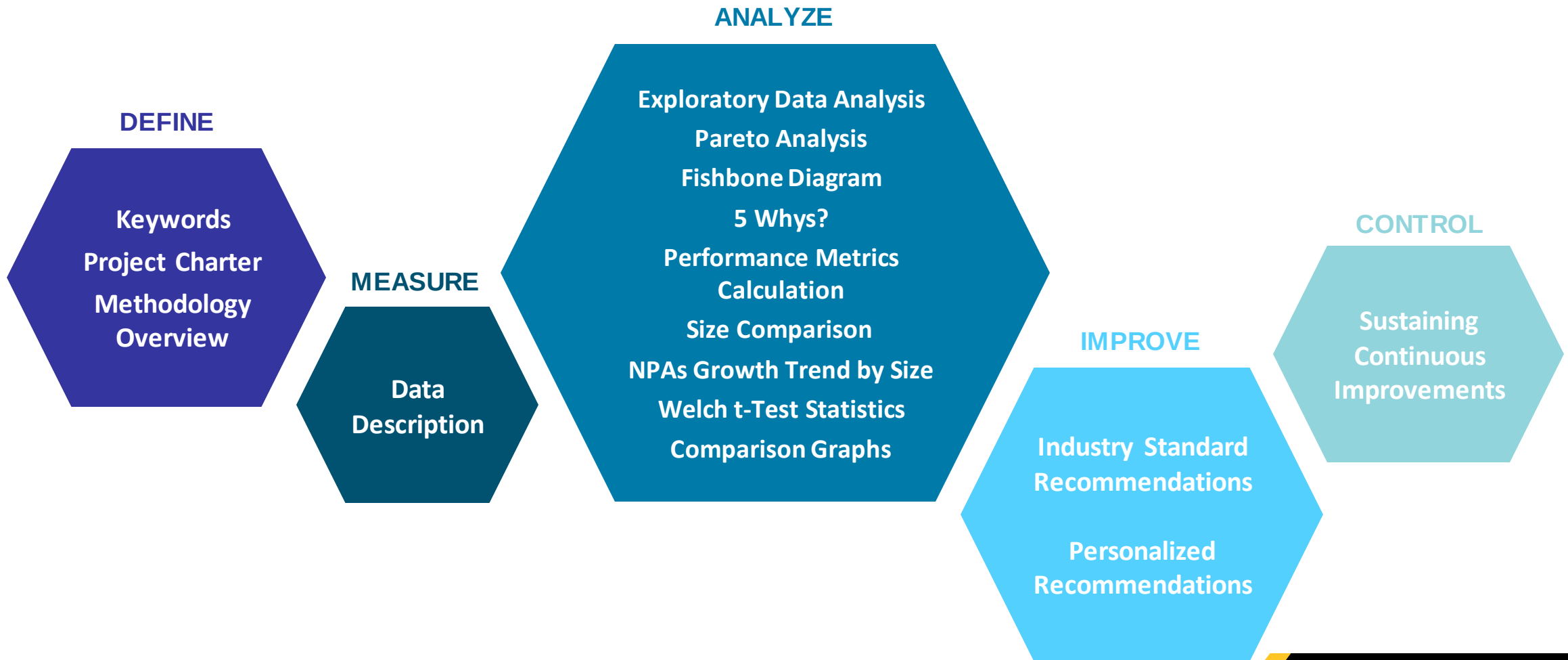

Does the Matthew Effect Exist for NIB's NPAs?

Lean Six Sigma Green Belt Certification Presentation Compiled by:
Min, Kan, Darren, Bhumit
MSBA Spring 2024

Agenda



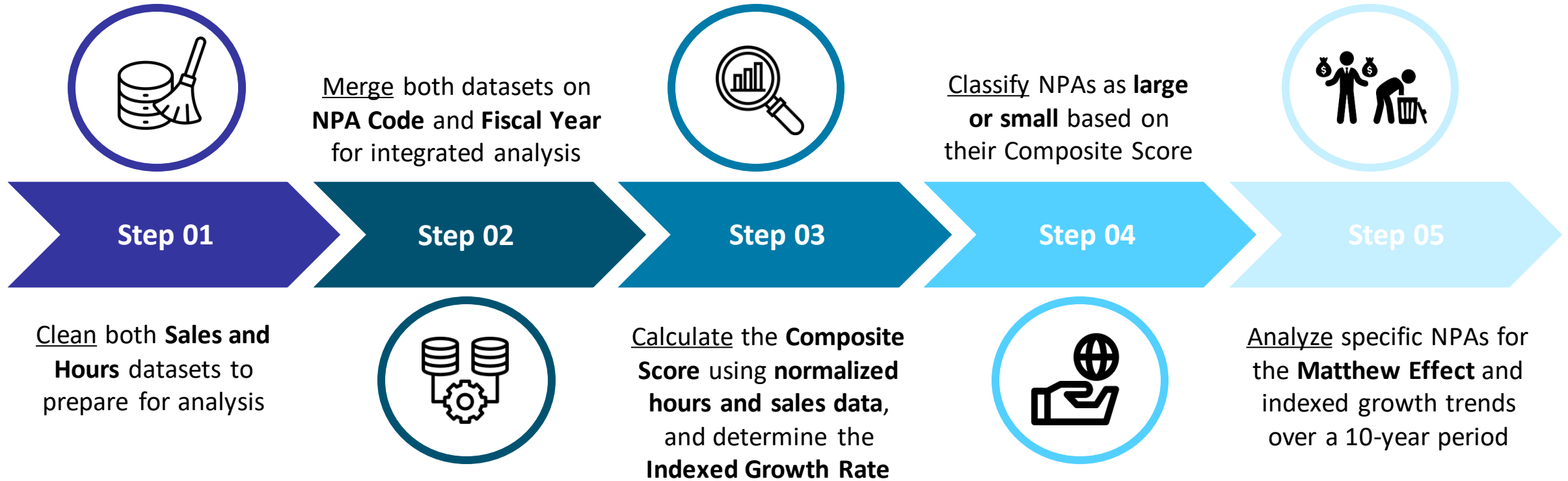
Keywords

NPA/NPA Code	A Nonprofit Agency that employs blind individuals to supply goods and services to the government under the AbilityOne Program, identified by a unique three-letter code.
Ability One Program	Federal effort that mandates government purchases from nonprofits employing blind or significantly disabled individuals to foster their employment.
JWOD Products/Services	Goods and services provided under the Javits-Wagner-O'Day Act, now part of the AbilityOne Program
DBLHs	Direct Blind Labor Hours, measure the work hours contributed by blind employees in producing goods or services under the AbilityOne Program.
Hours AllBlind	The total hours worked by all blind employees at a nonprofit agency, encompassing all their work activities, not just those under specific programs.
Hours AllSighted	The total hours worked by all sighted employees at a nonprofit agency, covering all their employment activities.
LoB/Sub-LoB	Line of Business and Sub-Line of Business classify the specific areas of work or services provided by an agency, detailing the main category and its subdivisions.

Project Charter

<u>Problem Statement</u> ❖ While many NPAs within the NIB network have seen growth, others have not, leading to a significant disparity. This potential Matthew Effect raises the question of whether larger NPAs continue to grow at the expense of smaller ones, which could have implications for blind employment growth and NIB policies.	<u>Executive Sponsor</u> Kenny Bao Director, Operations Support and Analysis, National Industries for the Blind (NIB)
<u>Objective and Scope</u> ❖ To analyze growth patterns among NIB's NPAs to determine the presence of the Matthew Effect, understand its causes, assess its impact on blind employment, and develop strategic recommendations for promoting equitable growth and employment opportunities.	<u>Process Stakeholders</u> ❖ NIB Leadership ❖ NPA Managers ❖ Blind Employees and Advocacy Groups ❖ Customers and Partners of NIB and NPAs
<u>Business Case and Benefit</u> ❖ Promoting balanced growth among NPAs is vital to prevent market monopolies and ensure the inclusivity of blind employment opportunities. Equitable policies could lead to a more diverse and resilient NIB network, benefiting all stakeholders.	

Methodology Overview



Data Description

DBLHs - Hours Data

Main Focus

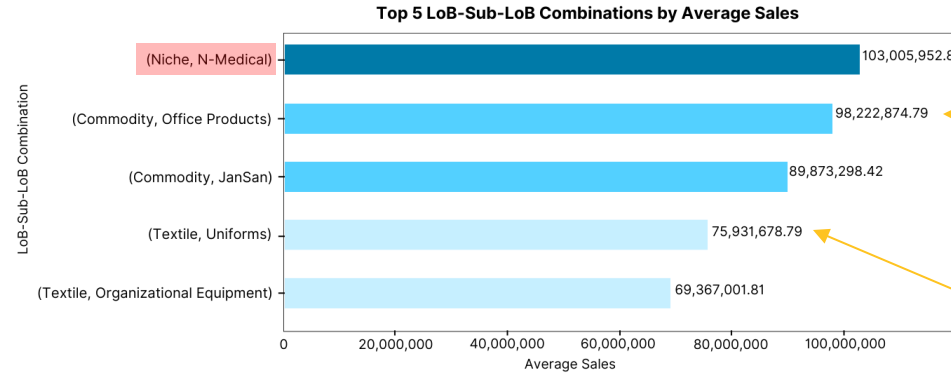
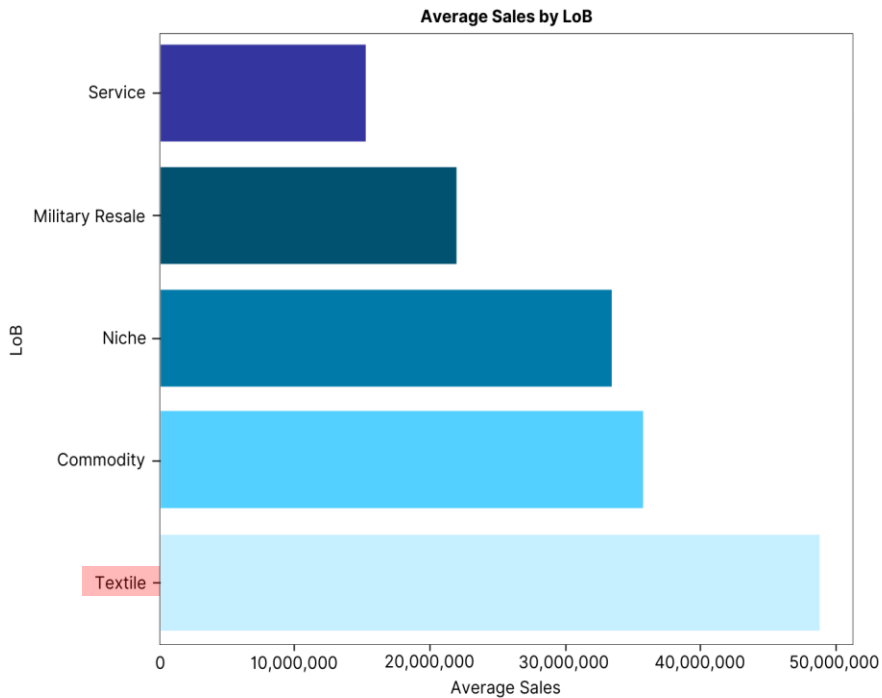
	Npa Code	Npa Name	Fiscal Year	Hours AllBlind JWOD Products	Hours AllBlind JWOD Services	Hours AllBlind JWOD Total	Hours AllSighted JWOD Products	Hours AllSighted JWOD Services	Hours AllSighted JWOD Total	Hours JWOD Products	Hours JWOD Services	Hours JWOD Total	Hours AllBlind JWOD Products Percentage	Hours AllBlind JWOD Services Percentage	Hours AllBlind JWOD Total Percentage
0	ALT	Alabama Industries for the Blind	2013	57,059.00	5,333.00	62,392.00	18,040.00	481.00	18,521.00	75,099.00	5,814.00	80,913.00	0.76	0.92	0.77
1	ARL	Arkansas Lighthouse for the Blind	2013	27,596.00	0.00	27,596.00	8,824.00	0.00	8,824.00	36,420.00	0.00	36,420.00	0.76	0.00	0.76
2	ORT	Tower for the Blind	2013	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	WVW	The Seeing Hand Association, Inc.	2013	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	TXU	New Vision Industries	2013	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

AbilityOne - Sales Data

	LoB	Sub-LoB	NPA Code	NPA Name	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
0	Commodity	C- Food Service	ALT	ALABAMA INDUSTRIES FOR THE BLIND	10,320.00	11,446.57	5,035.11	435.46	NaN	299.67	NaN	128.43	8,390.76	1,369.92	8,441.20	NaN	NaN	NaN
1	Commodity	C- Food Service	GAB	GEORGIA INDUSTRIES FOR THE BLIND	320,166.00	481,833.00	377,426.43	419,364.00	34,128.00	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	Commodity	C- Food Service	LAN	Lighthouse Louisiana	29,863,862.30	10,892,140.89	5,601,024.22	7,287,336.48	9,475,066.28	11,288,858.75	12,396,631.48	14,788,292.98	10,768,413.11	12,709,563.45	10,023,489.53	11,271,026.39	10,921,114.40	2,577,868.19
3	Commodity	C- Food Service	NCD	LC INDUSTRIES, INC	26,300,707.55	15,735,884.61	4,171,347.77	2,862,695.91	5,060,876.24	9,573,563.36	12,076,760.92	11,216,185.07	8,782,975.68	9,863,109.84	8,303,815.10	9,935,683.90	10,389,077.20	2,743,466.77
4	Commodity	C- Food Service	NJW	BESTWORK INDUSTRIES FOR THE BLIND, INC	103,468.37	92,433.30	17,429.10	16,418.58	29,242.42	16,559.90	45,284.56	34,702.85	28,375.62	56,315.35	33,893.86	51,772.00	24,084.00	2,420.00

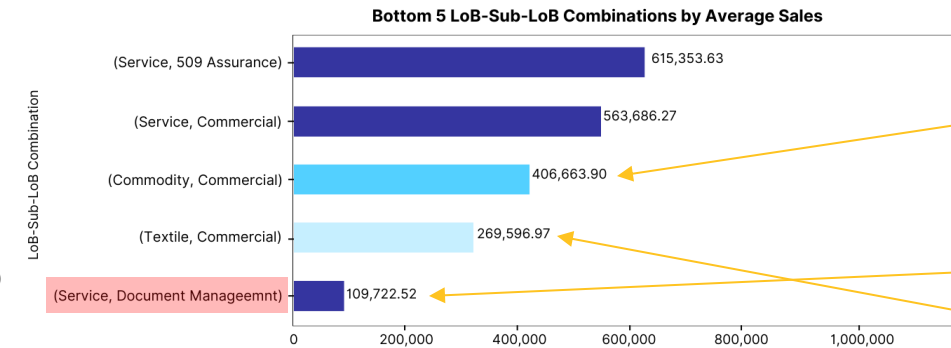
Exploratory Data Analysis – *AbilityOne*

There isn't a Sub-LoB for Military Resale LoB



Sub-LoB with the Largest Average Sales in Each LoB

LoB	Sub-LoB	Average Sales
Commodity	Office Products	98,222,874.79
Military Resale	Military Resale	22,005,162.15
Niche	N-Medical	103,005,952.82
Service	Supply Chain	47,708,839.13
Textile	Uniforms	75,931,678.79



Sub-LoB with the Smallest Average Sales in Each LoB

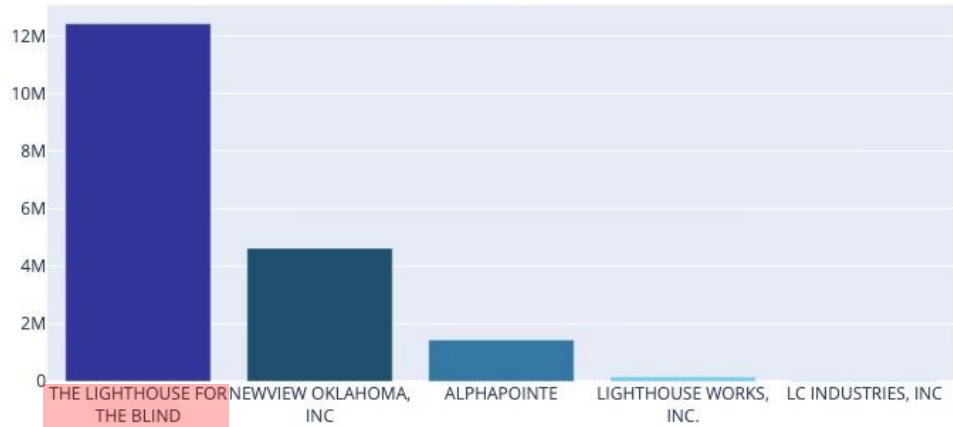
LoB	Sub-LoB	Average Sales
Commodity	Commercial	406,663.90
Military Resale	Military Resale	22,005,162.15
Niche	N-Personal Safety	2,252,168.96
Service	Document Management	109,722.52
Textile	Commercial	269,596.97

- ❖ **LoB:** Over the 13-year period, **Textile** LoB maintained the largest average sales
- ❖ **Sub-LoB:** Tables on the right side indicate the largest/smallest Sub-LoB under each LoB
- ❖ **LoB-Sub-LoB Combination:** Within the top/bottom 5 LoB-Sub-LoB combinations, **N-Medical** products under **Niche** LoB generated the highest average sales, while **Document Management** under **Service** LoB generated the lowest average sales

Products in the Niche LoB generates higher sales, leading to significantly higher average sales compared to other categories within this group

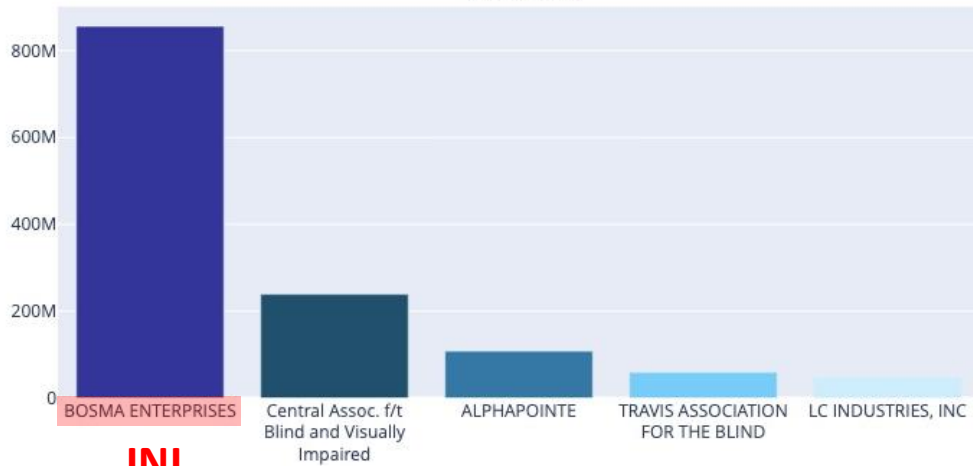
Exploratory Data Analysis – *AbilityOne*

Commercial
C-Medical



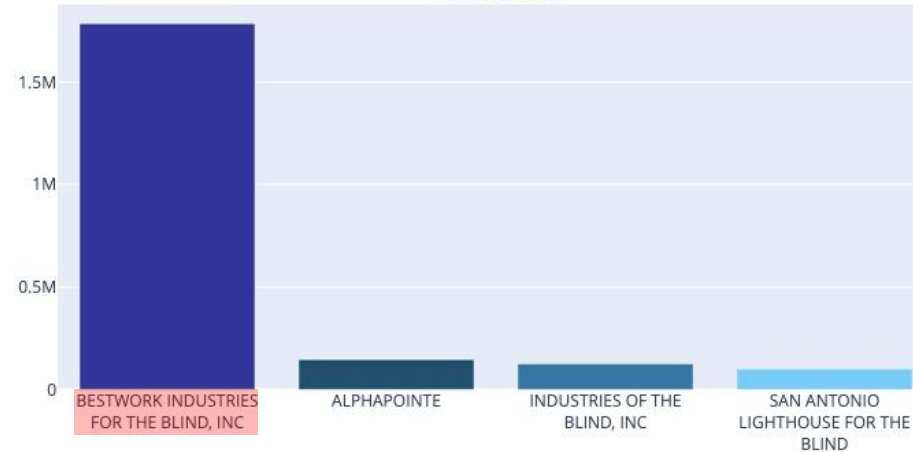
MOS

Niche
N-Medical



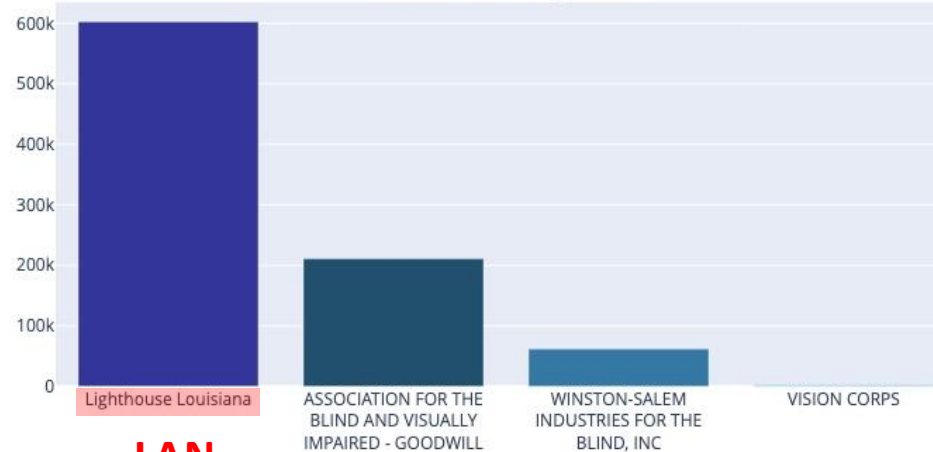
INI

Textile
Commercial



NJW

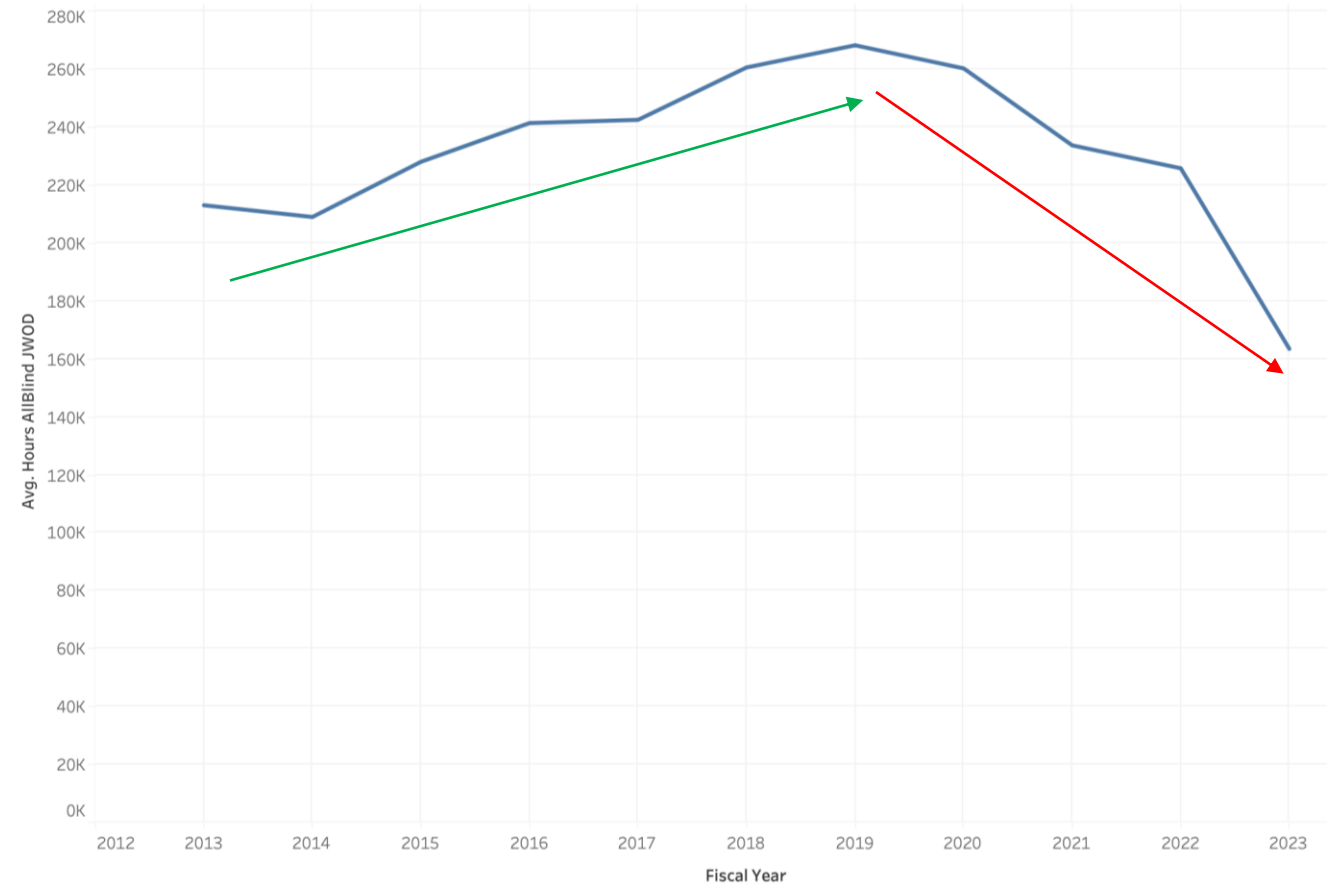
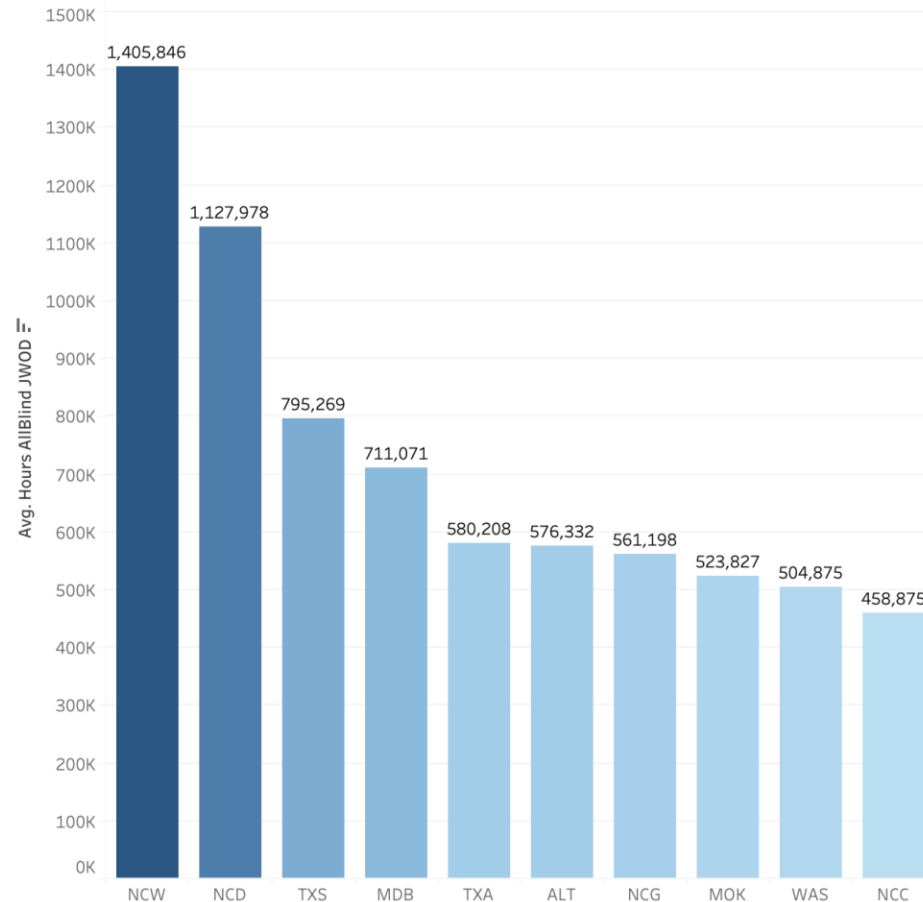
Service
Document Management



LAN

- ❖ We analyzed the **top 5 NPAs** contributing to each LoB-Sub-LoB combination, ranked by total sales
- ❖ This observation suggests the **potential for market concentration** within specific LoB-Sub-LoB combinations

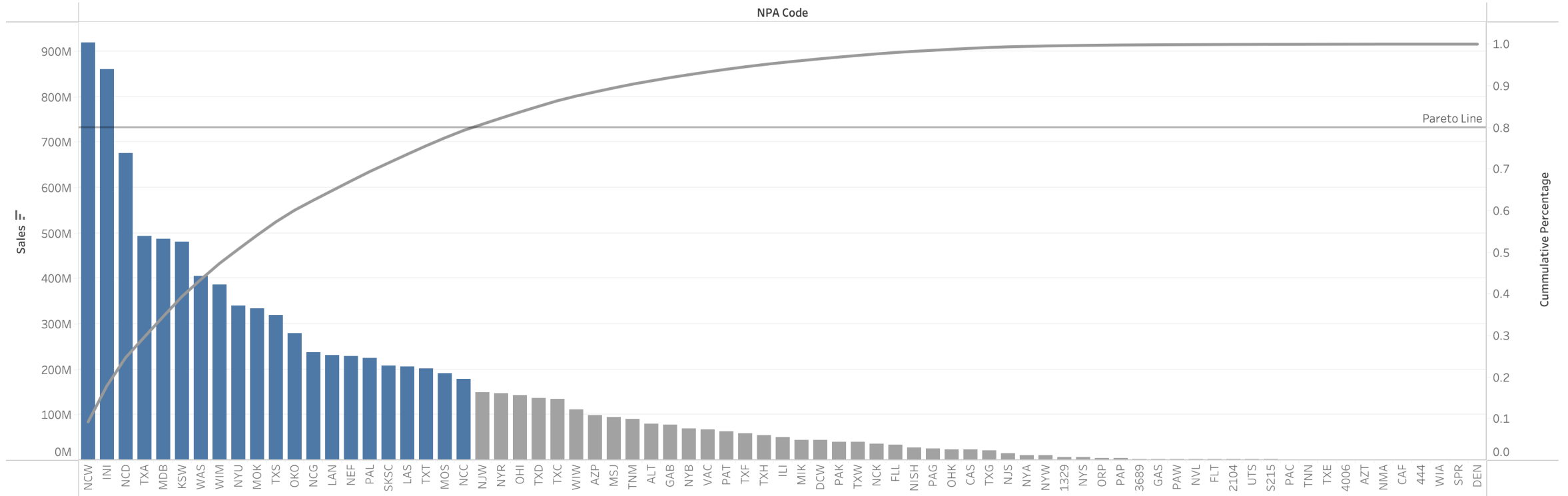
Exploratory Data Analysis – *DBLHs*



- ❖ **Bar Chart:** Significant variation in *Avg. Hours AllBlind JWOD* with larger NPAs like NCW suggests a need to analyze the correlation between NPA size and allocated hours
- ❖ **Line Chart:** The rise and subsequent decline in *Avg. Hours AllBlind JWOD* until 2019 hints at underlying cyclical shifts, warranting further examination

Pareto Analysis

Pareto Analysis 80/20



This Pareto analysis illustrates that a small number of NPA Codes generate the majority of sales, with the left-most NPA representing the top-performing agencies. The cumulative percentage line indicates that focusing on these key areas could yield significant revenue impact, exemplifying the 80/20 rule in sales concentration. For our further analysis, we will focus on these NPA for Mathew Effect as well.

We see that 80% of total sales are made from 25% of the companies.

Fishbone Diagram



5 Whys?

Problem: Certain NPAs continue to grow, while others are either eliminated, absorbed, or remain small

WHY?

Market Dominance by Select NPAs Reducing Competitive Opportunities for Others

WHY?

Limited Manpower (DBLH), Insufficient Availability of Direct Blind Labor Hours Restricting Operational Capacity

WHY?

Scarce Resources (Materials), Challenges in Procuring Essential Production Materials Impacting NPA Operations

WHY?

Diversified LoB / Sub-LoB, Advantages of an Expanded and Specialized Line of Business Portfolio in NPA Growth Dynamics

WHY?

Limited Opportunities to Grow Business Through JWOD for Smaller NPA

Methodology: Performance Metrics Calculation



STEP 1

Normalize AbilityOne Sales and DBLHs Hours data using

$$Z = (X - \mu) / \sigma$$



W1: Normalized AbilityOne Sales



W2: Normalized DBLHs (JWOD Total Hours)



STEP 2

Calculate Composite Score for each NPAs, annually

$$CS = 0.5 * W1 + 0.5 * W2$$



STEP 3

Track Indexed Growth Rate annually from a 2013 base year

$$IG = (\Delta \text{ Composite Score} / \text{Base Composite Score}) * 100$$

Analyze the year-wise trend of Indexed Growth over a 10-year period for Mathew Effect

Performance Metrics Calculation

NPA Code	FY	Sales	Hours AllBlind JWOD	Normalize Hour All Blind JWOD Total	Normalize Total Ability One Sales	Composite Score	Indexed Growth
ALT	2013	8166494.632	638320	1.437553919	-0.288326967	0.574613476	100
ALT	2014	6430416.154	611759	1.343736703	-0.392549945	0.475593379	82.76752969
ALT	2015	5911198.886	617862	1.365293363	-0.423720407	0.470786478	81.93098458
ALT	2016	6368979.9	641426	1.448524751	-0.396238182	0.526143284	91.56473121
ALT	2017	5325098.835	667828	1.541780356	-0.458906082	0.541437137	94.22632085
ALT	2018	4755448.49	607675	1.329311436	-0.493104222	0.418103607	72.76258298
ALT	2019	5399844.86	596086	1.288377446	-0.454418812	0.416979317	72.56692276
ALT	2020	3084847.94	567841	1.188612103	-0.593396326	0.297607889	51.79270958
ALT	2021	4606722.72	403318	0.607493552	-0.50203276	0.052730396	9.176672406
ALT	2022	5382625.96	611538	1.342956099	-0.455452523	0.443751788	77.22613665
ALT	2023	5567267.65	376003.5	0.511014868	-0.444367825	0.033323521	5.799293391
AZP	2016	6841158.359	281236	0.176282628	-0.367891626	-0.095804499	-231.9564204
AZP	2017	6327096.392	258469	0.095866358	-0.398752598	-0.15144312	-366.6654942
AZP	2018	6560770.75	228379	-0.010415786	-0.384724294	-0.19757004	-478.3453767
AZP	2019	6309892.91	232761	0.005062059	-0.399785384	-0.197361663	-477.8408658
AZP	2020	6804036.45	236789	0.019289525	-0.370120187	-0.175415331	-424.705651
AZP	2021	6618244.19	161471	-0.246744324	-0.381273958	-0.314009141	-760.261125
AZP	2022	7230513.6	224022	-0.025805328	-0.344517244	-0.185161286	-448.3020048
AZP	2023	7643690.72	209491	-0.077130879	-0.319712749	-0.198421814	-480.4076437
AZP	2013	10011724.18	304982	0.260156865	-0.177551263	0.041302801	100
AZP	2014	7684376.382	294313	0.222472445	-0.317270244	-0.047398899	-114.7595271
AZP	2015	8055888.468	269668	0.135422813	-0.294967049	-0.079772118	-193.1397292

STEP 1. Normalization of Data

- To adjust for different scales and variability & to assess meaningful comparison across years and NPAs
- Normalize Hour All Blind JWOD Total** = $\text{Hour AllBlind JWOD} - \text{AVERAGE}(\text{Hour AllBlind JWOD}) / \text{STDEV.P}(\text{Hour AllBlind JWOD})$
- Normalize Total Ability One Sales** = $\text{Sales} - \text{AVERAGE}(\text{Sales}) / \text{STDEV.P}(\text{Sales})$

Performance Metrics Calculation

NPA Code	FY	Sales	Hours AllBlind JWOD	Normalize Hour All Blind JWOD Total	Normalize Total Ability One Sales	Composite Score	Indexed Growth
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STEP 2. Composite Score Calculation

- It combines normalized measures of hours and sales into a single metric to evaluate NPA's performance
- Composite Score** = **Normalized Hour All Blind JWOD Total** * 0.5 + **Normalized Total Ability One Sales** * 0.5

STEP 3. Indexed Growth Rate

- It reflects the relative change in performance year over year, with the base year indexed at 100
- 100 + (Change in Composite Score / Base Year Composite Score) * 100**
- Change = Current Year Composite Score – Base Year Composite Score**

Size Comparison

Large NPAs were top 33 percentile and Small NPAs were bottom 33 percentile. We then plotted Indexed growth with time for each NPA.

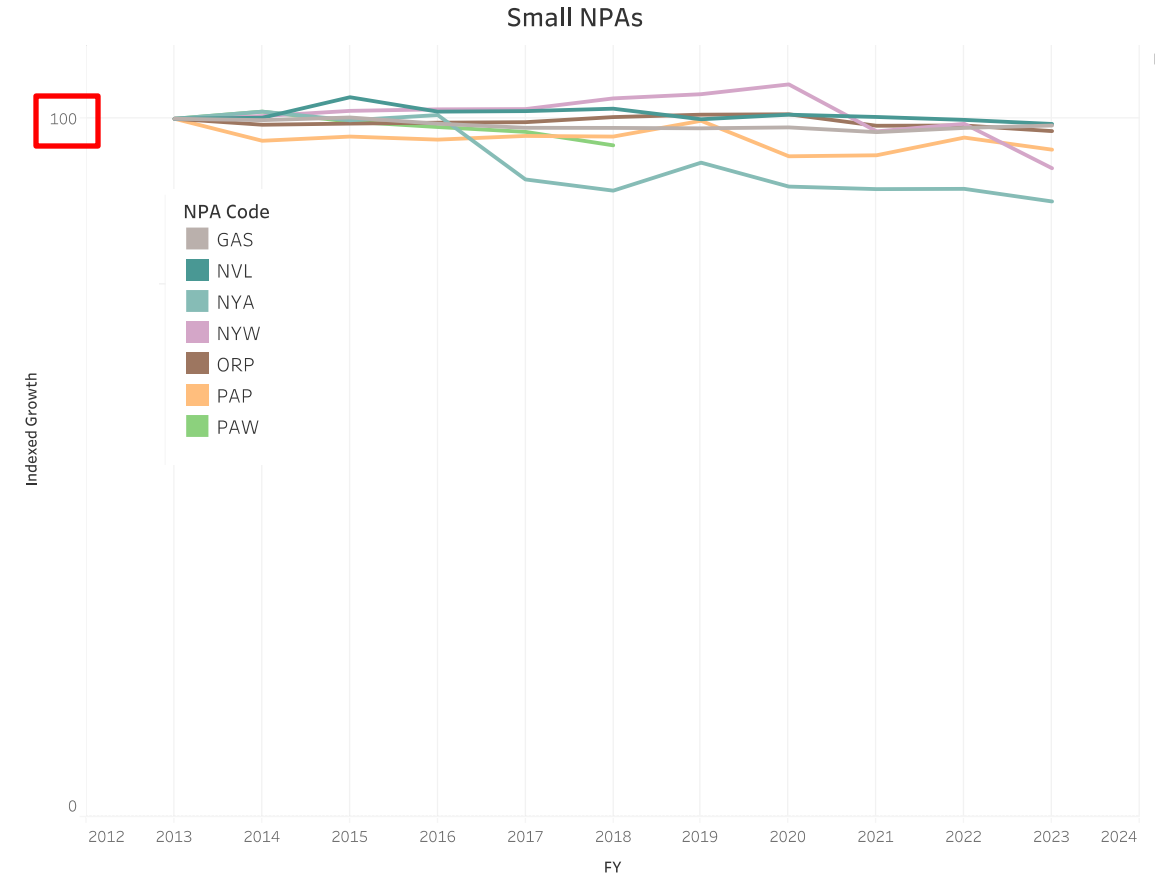
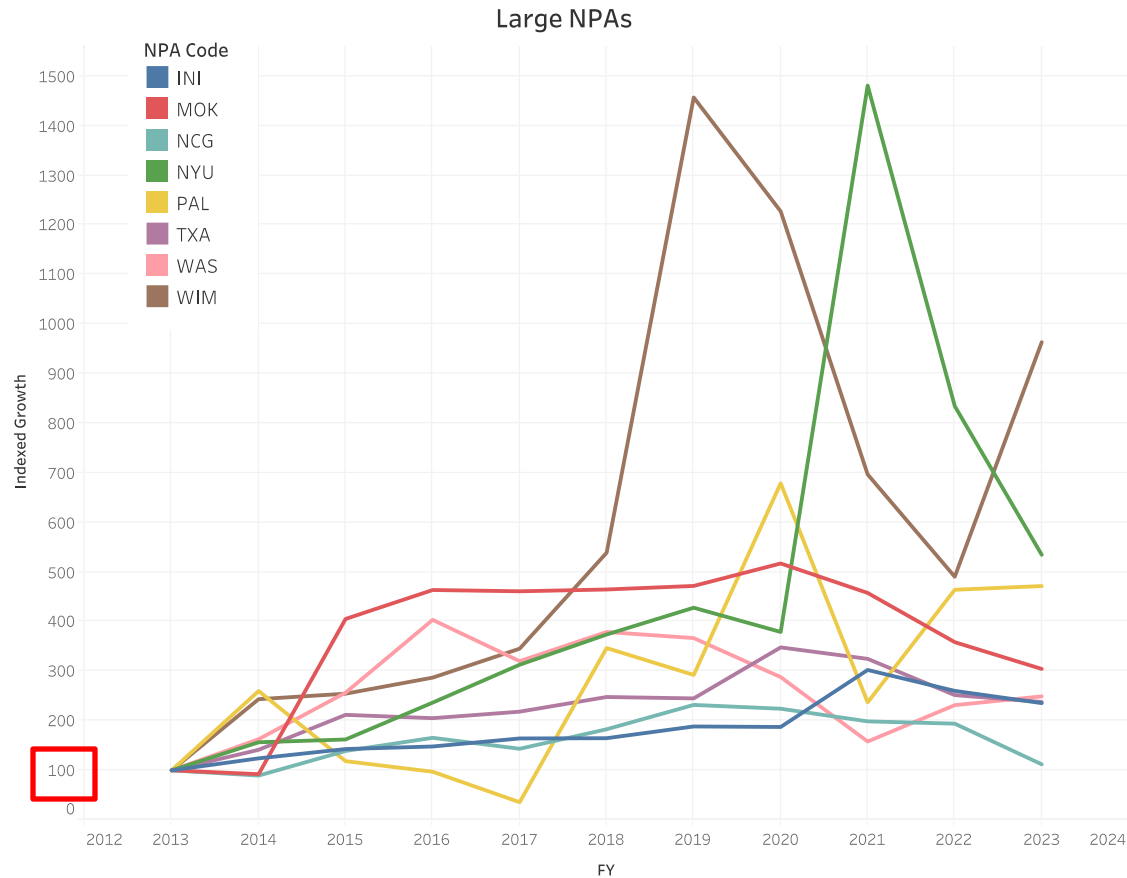
Below are the NPA's that were **large in 2013** and got **larger** with time
In respect to the indexed growth plotted vs 10 year period. (9/17)

<u>NPA</u>	<u>Line of Business</u>	<u>Sub Line of Business</u>
INI	Niche	N-Medical
LAS	Commodity	Office Products
MOK	Commodity, Niche, Textile, Service	Office Products, N-Medical, Org. Clothing, Teleservices
NCG	Commodity, Service, Textile	Office Product, Supply Chain, Org. Clothing, Org. Equipment, Uniforms
NYU	Niche, Service, Textile, Commodity	N-Medical, Teleservices, Org. Clothing, Office Prodcuts
PAL	Textile, Service, Niche	Org. Equipment, Supply Chain, N-Food Services, Hazardous Material
TXA	Niche, Service, Textile	N-Medical, Administration, Supply Chian, Org. Equipment, Org Clothing
WAS	Commodity, Niche, Textile	Office Products, N-Food Services, N-Medical, Org. Equipment
WIM	Commodity, Niche	Office Product, Industrial, N-Medical

Below are the NPA's that were **small in 2013** and got **smaller** with time. (7/20)

<u>NPA</u>	<u>Line of Business</u>	<u>Sub Line of Business</u>
GAS	Service	Administration
NVL	Commodity	Office Products
NYA	Niche, Textile	Industrial, Hazardous Material, Org. Clothing
NYW	Service	Teleservices
ORP	Textile	Org. Clothing
PAP	Niche, Commodity	N-Food Services, Food Servies
PAW	Niche, Textile	N-Medical, Org. Equipment

NPA's Growth Trends by Size



- ❖ **Base Year:** 2013 for both charts
- ❖ **Left Chart:** Shows growth in Large NPAs over 10 years
- ❖ **Right Chart:** Shows no growth or decline in Small NPAs over 10 years

Welch t-Test Statistics

Definition

A t-test compares the **mean** of two groups and determines whether they are **statistically different** from each other.

Null Hypothesis

There is **no difference** in Indexed Growth Rates between Large and Small NPAs

Alternative Hypothesis

There is **a difference** in Indexed Growth Rates between Large and Small NPAs.

Calculation

It calculates the **t-statistic**, which measures the size of the difference relative to the variation in your data.

T-statistic = 8.79

P Value = 2.50e-13

Significance Value = 0.05 (95%)

Assessment

Hence to formally assess whether the size change is statistically significant, we performed t-test, comparing the two groups, **large and small NPAs**.

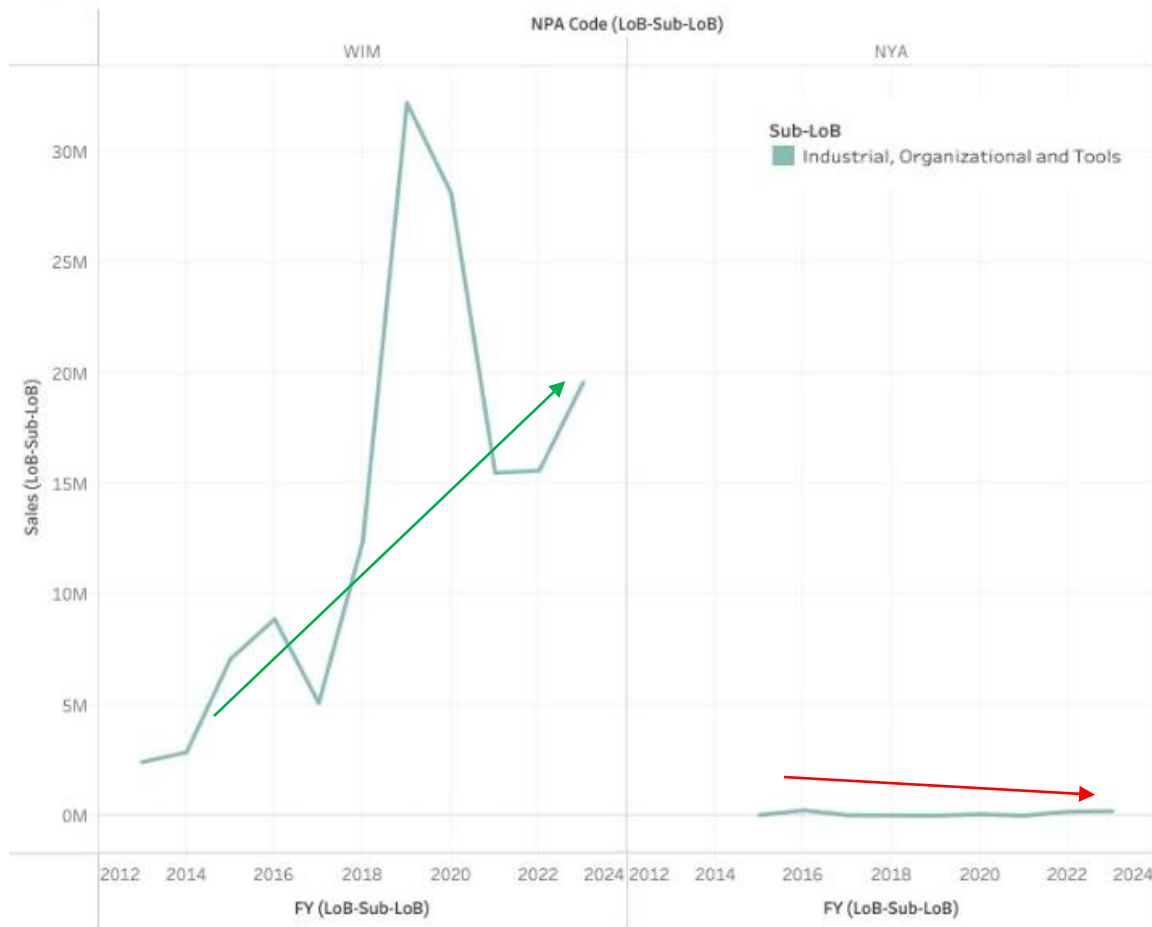
- ❖ As **p-value < significance value**, we reject the null hypothesis, indicating a statistically significant difference in growth rates between large and small NPAs.
- ❖ The test suggests that **larger NPAs exhibit notably higher growth rates than smaller ones**, possibly reflecting the Matthew Effect

Comparision Graph Showing Mathew Effect

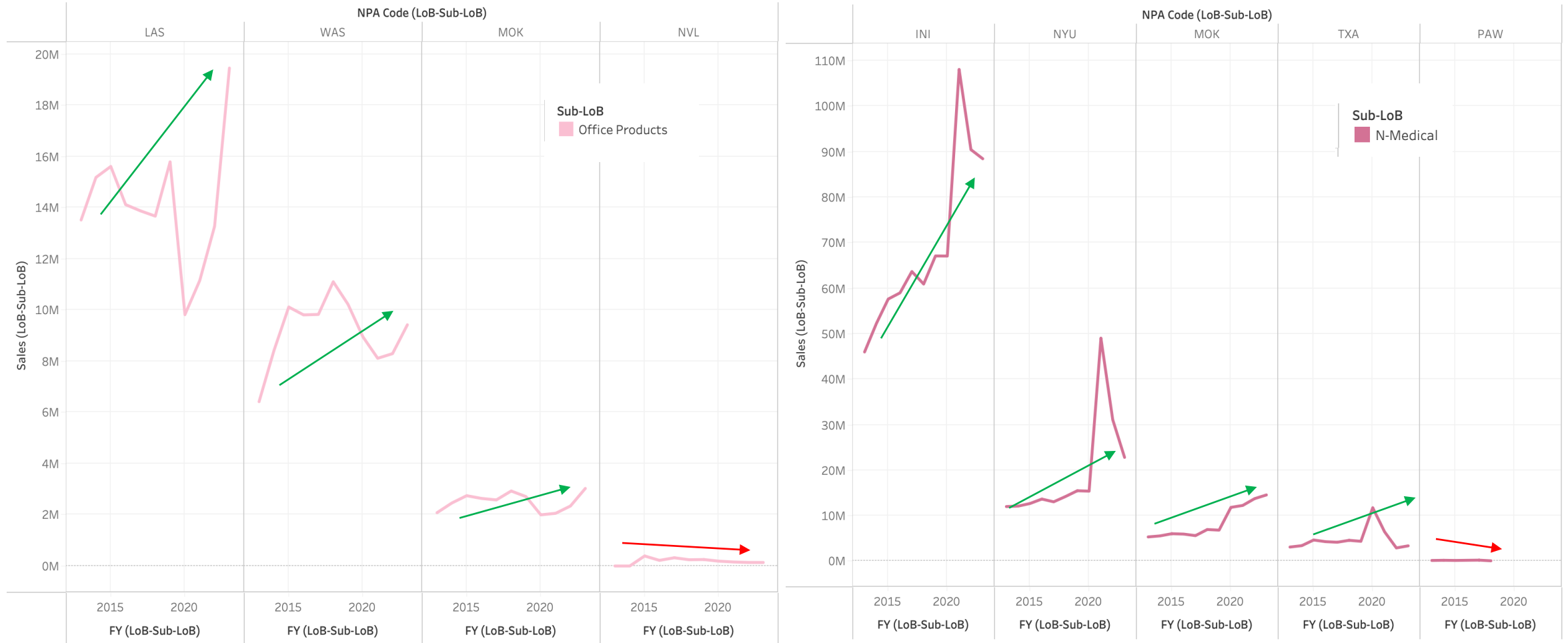


- ❖ To compare the NPAs at a granular level of Sub-LoB, we only had data for sales. Therefore, we analyzed them to determine if the t-test statistic provides the correct interpretation or not.
- ❖ From the previous slide, we observed that not all LoB-Sub-LoB combinations exhibit the Matthew Effect; only a few of them do in terms of sales.
- ❖ More data in terms of hours, provided at the granular level of LoB-Sub-LoB, can help us analyze further.

Comparision Graph Showing Mathew Effect



Comparision Graph Showing Mathew Effect



Industry Standard Recommendation

❖ Training and Development

- Roll out skill-enhancement programs tailored for smaller NPAs to boost their market competitiveness.
- Introduce leadership development programs designed to strengthen strategic planning and execution within smaller NPAs.

❖ Innovation and Technology

- Assist smaller NPAs in embracing innovative technologies that can optimize their operations.
- Offer grants for projects that drive innovation, helping smaller NPAs to break new ground.

❖ Performance Monitoring

- Set up a framework for ongoing tracking and assessment of NPA performance, identifying areas for targeted support.
- Facilitate benchmarking initiatives, allowing smaller NPAs to learn from the successes of others.

❖ Contract Assignment Policy Review

- Review whether current Ability One Contract Assignment policy may be inadvertently creating the Matthew Effect in favor of incumbent NPAs

Personalized Recommendation

❖ Resource Allocation

- Implementing a funding model that prioritizes equitable growth, channeling more resources to our Smaller NPAs to help them expand their footprint ***without sacrificing employment of Blind workers at Large NPAs***
- Launch investment initiatives focused on building the infrastructure and operational capacity of smaller NPAs.

❖ Collaboration and Partnerships

- Foster a culture of partnership, encouraging knowledge exchange and shared growth strategies between large and small NPAs.
- Support and incentivize joint projects that can empower smaller NPAs to participate in larger bids and contracts.

❖ Networking and Visibility

- Increase the visibility of smaller NPAs through dedicated showcases and exposure to a broader audience.
- Organize networking events that can open doors to new business opportunities for our smaller NPAs.

❖ Diversification

- Guide smaller NPAs toward broadening their service and product offerings, reducing dependency on limited market segments and enhancing resilience.

Sustaining Continuous Improvement

❖ Development of Monitoring Systems

- Create SOPs for any new processes to address and monitor the Matthew Effect.
- Set automated alerts for anomalies indicating potential Matthew Effects.

❖ Regular Review Meetings

- Institutionalize periodic performance review sessions.
- Discuss data-driven insights and effectiveness of interventions in these sessions.

❖ Adjustment of Strategies

- Set up a structured feedback system for continuous process improvement.
- Refine strategies based on real-time data and feedback.

❖ Audit and Compliance Checks

- Conduct regular audits to verify data integrity and process adherence.
- Implement compliance checklists for process verification.

Thank you
